```
54Fe(d,p)55Fe 16 MeV (ADWA in/ KD out), 0.411 MeV p0.5
0.10    40.0    0.20    0.10    5.0     -0.3
 00. 20.   +.00    F F
0 10.        55.    0.1  1
0.0     1    1 1 0  48
 1 1 0 0 1 3 0 0 0 1 0 0 1
2H      2.0141  1.0         1  54Fe    53.9396 26.0    0.0000
1.0    +1 0.0               1  0.0    +1 0.00
1H      1.0078  1.0         1  55Fe    54.9383 26.0    7.0730
0.5    +1 0.0               2  0.5    -1 0.411

  1 0  0     54.0     0.0   1.268
  1 1  0    104.5   1.197   0.702    1.22    1.197    0.702
  1 2  0                            14.98    1.282    0.584
  1 3  0    11.31   1.014   0.621   -0.12    1.014    0.621
  2 0  0     55.0     0.0    1.28
  2 1  0    53.08   1.197   0.670   1.282    1.197    0.670
  2 2  0                             8.15    1.282    0.547
  2 3  0     5.54   1.015    0.59   -0.07    1.015    0.590
  3 0  0     1.00            1.25
  3 1  5     1.00            1.00
  3 3  5     1.00            1.00
  3 4  5     1.00            1.00
  3 7  5     1.00            1.00
  4 0  0     55.0    0.0    1.25
  4 1  0     50.0    1.25    0.65
  4 3  0      6.0    1.25    0.65
0
  1   2 1 2-1 3   1 0 2 0.5 1 1.5 1   3  0  2.2246  0  3  0
  3     1 2-2 0   2 1   0.5   0.5     4  0  8.8872  1  0

  -2    1    7 0-1 0
        1    1    1   1  1.0000
       -2    1    1   3  1.0000
    0   1    1
16.0
EOF
```

(Just a title)

θ range and step

relative Q value from Kinace or whatever

Excitation energy

d

p

Optical potentials

A out

d binding energy

s

l

$j = |\vec{l} + \vec{s}|$

target-like reaction product binding energy

nodes (SHO n)

(So 2 1 0.5 0.5 ⇒ $2p_{1/2}$)